

MANIPAL UNIVERSITY JAIPUR

Department of Computer Science and Engineering

Course Hand-out

A. Basic Details:

Programme Name:	B. Tech
Course Name:	Software Engineering
Course Code:	CSE3102
LTPC (Lecture Tutorial Practical Credits):	[3 1 0 4]
Session:	2026-2027
Class:	III Year/ V Semester
Course Coordinator(s):	Dr. Akshay Jadhav and Dr. Susheela Vishnoi
Course Instructor(s):	Dr. Manmohan Sharma (Section A), Dr. Bali Devi (Section B), Dr Gireesh Kumar (Section D & H), Ms. Shweta Sharma (Section E), Dr. Prashant Vats (Section G), Dr. Vijay Hasanpuri (Section F & J), Ms. Surbhi Syal (Section I), Dr. Susheela Vishnoi (Section K), and Dr. Akshay Jadhav (Section C)

B. Introduction: The aim of this course is to enable a clear understanding and knowledge of the foundations, techniques, tools and processes in the area of software engineering and its practices followed in software industry. The course will prepare students to apply engineering methods and processes on software projects.

C. Course Outcomes:

CO Statement	CO	Bloom's Level	Target Attainment %	Target Attainment Level
Demonstrate an understanding of the fundamentals of software engineering principles, methodologies, and techniques.	CSE3102.1	L2 – Understand	80%	3
Illustrate software requirements analysis, Software Requirements Specification (SRS) documents, software metrics, and system modeling.	CSE3102.2	L2 – Understand	70%	3
Develop software systems to meet specified requirements efficiently.	CSE3102.3	L3 – Apply	70%	2
Analyze test cases for software systems using various testing techniques.	CSE3102.4	L4 – Analyze	70%	2
Assess software quality, analyze evolutionary processes, and evaluate software security measures effectively.	CSE3102.5	L5 – Evaluate	70%	3

Here, the level indicates the cognitive level of Revised Bloom's Taxonomy.

Information about attainment levels:

Attainment (%)	Level
< 60 %	0
≥ 60% < 70%	1
≥ 70% < 80%	2
≥ 80	3

D. Program Outcomes and Program Specific Outcomes:

[PO.1]. **Engineering knowledge:** Apply the knowledge of basic science and fundamental

computing in solving complex engineering problems

[PO.2]. Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

[PO.3]. Design/development of Computing solutions: Design solutions for complex IT engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the Information oriented public health and safety, and the cultural, societal, and environmental considerations

[PO.4]. Conduct investigations of complex problems: Use IT domain research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions

[PO.5]. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations

[PO.6]. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice

[PO.7]. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development

[PO.8]. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practices

[PO.9]. Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse IT teams, and in multidisciplinary settings.

[PO.10]. Communication: Communicate effectively on complex computing engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions

[PO.11]. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments

[PO.12]. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

[PSO1]: Design and develop optimized computational solutions using appropriate programming paradigms, contributing to sustainable development to ensure efficient resource utilization.

[PSO2]: Create intelligent or innovative solutions to real-world challenges by applying modern tools, software engineering practices, and emerging technologies.

[PSO3]: Process and communicate data effectively while ensuring software security, data privacy, and reliable data communication in practical applications.

E. Assessment Plan:

Criteria	Description		Maximum Marks
Internal Assessment (Summative)	Mid-Term Examination (Close Book)		30
	CWS	Attendance	05
		Assignments (Total 2)	10
		Quiz (Total-3, 10 Marks each)	15
End Term Exam (Summative)	End Term Exam (Close Book)		40
	Total		100

Attendance (Formative)	A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking the End Semester examination. The allowance of 25% includes all types of leaves including medical leaves.
Make up Assignments (Formative)	Students who miss a class will have to report to the teacher about the absence. A makeup assignment on the topic taught on the day of absence will be given which has to be submitted within a week from the date of absence. No extensions will be given on this. The attendance for that day of absence will be marked blank, so that the student is not accounted for absence. These assignments are limited to a maximum of 5 throughout the entire semester.
Homework/ Home Assignment/ Activity Assignment (Formative)	There are situations where a student may have to work in home, especially before a flipped classroom. Although these works are not graded with marks. However, a student is expected to participate and perform these assignments with full zeal since the activity/ flipped classroom participation by a student will be assessed and marks will be awarded.

F. Syllabus:

Module No.	Module Title	Topics Covered
1	Introduction to Software Engineering	The Evolving Role of Software, The Changing Nature of Software, Legacy Software, Software Myths, Software Engineering as a Layered Technology, Process Framework, Capability Maturity Model Integration (CMMI).
2	Software Process Models & Agile Development	Specialized Process Models, Unified Process, Agile Process Models, Agile Principles, Adaptive Software Development.
3	Software Engineering Practices & System Engineering	Communication Practice, Planning Practices, Modeling Practices, Construction Practice, Deployment, Computer-Based Systems, The System Engineering Hierarchy, Business

		Process Engineering, Product Engineering.
4	Modeling, Analysis and Quality Management	Data Modeling Concepts, Object-Oriented Analysis, Flow-Oriented Modeling, Taxonomy of Quality Attributes, Perspectives of Quality, Quality System, Software Quality Assurance (SQA), Capability Maturity Model.
5	Project Management & DevOps	Estimation and Project Planning Process, Software Scope and Feasibility, Human Resources, Empirical Estimation Models, Introduction to DevOps, Cloud Computing & Virtualization, Migration to DevOps, DevOps Tools.

Software Engineering Syllabus: Introduction: The Evolving Role of Software, The changing nature of software, Legacy software, Software Myths. Software Engineering: A Layered Technology, a Process Framework, the Capability Maturity Model Integration (CMMI), Specialized Process Models, and the Unified Process. Agile development: Agile Process Models Software Engineering Practice, Communication Practice, Planning Practices, Modeling Practices, Construction Practice, Deployment Computer-Based Systems, The System Engineering Hierarchy, Business Process Engineering: An Overview. Product Engineering: An Overview, Data Modeling Concepts, Object Oriented Analysis, Flow-Oriented Modeling, Taxonomy of Quality Attributes, Perspectives of Quality, Quality System, Software Quality Assurance, Capability Maturity Model Observation on Estimation, The Project Planning Process, Software Scope and Feasibility, Human Resources, Empirical Estimation Model, **Introduction to DevOps, Cloud Computing And Virtualization, Migration to DevOps, DevOps Tools.**

Reference Book:

1. R. Pressman, Software Engineering: A Practitioners Approach, (8e), McGraw Hill Pubs, 2019.
2. M. Walls, Building a Dev Ops Culture, O'Reilly Publications, 2013.
3. J. Joyner, Dev Ops for Beginners, Dev Ops Software Development Method guide for software developers and IT professionals, Mihails Konoplovs, 2015.

G. Lecture Plan:

Lecture	Topics	Session Outcome	Mode of Delivery	Corresponding CO	Mode of Assessing CO
1	Introduction to software engineering	To introduce the significance of software engineering	Lecture	CSE3102.1	Mid Term, Quiz, Assignment & End Term
2	Software, Emergence of software engineering, evolving role of software, The changing nature of software.	Define role and nature of software	Lecture	CSE3102.1	Mid Term, Quiz, Assignment & End Term
3	Legacy software, Software Myths, Software crisis.	Analyze types of software and myths associated with them.	Lecture	CSE3102.1	Mid Term, Quiz, Assignment & End Term
4	Introduction to software engineering: Layered Approach, Generic approach, Process Framework.	Recall software engineering layered approach and process framework	Lecture	CSE3102.1	Mid Term, Quiz, Assignment & End Term
5	The process, software products, Software Characteristics, Applications.	Recall software products, Software Characteristics, Applications.	Lecture	CSE3102.1	Mid Term, Quiz, Assignment & End Term
6	Process Models- Conventional Process Model (Traditional Waterfall Model, prototype, RAD)	Appraise software development life cycle, Assessment Model-CMMI	Lecture	CSE3102.1	Mid Term, Quiz, Assignment & End Term
7	Process Models- Conventional Process Model (Traditional Waterfall Model, prototype, RAD)	Choose between process Models	Lecture	CSE3102.1	Mid Term, Quiz, Assignment & End Term
8	Evolutionary Process Model (Incremental Model, Spiral models, Component based process model, unified process model), Comparison of various models	Compare Evolutionary Process Model	Lecture	CSE3102.1	Mid Term, Quiz, Assignment & End Term
9	An Agile view of process, Human Factors	Define agile view of process	Lecture	CSE3102.1	Mid Term, Quiz, Assignment & End Term
10	Agile Process Models (Extreme programming, Adaptive Software	Apprise Agile process models and its adaptive	Lecture	CSE3102.1	Mid Term, Quiz, Assignment

	Development, Dynamic System Development method)	software development using Extreme programming			& End Term
11	Agile process Models (Scrum, Crystal, Feature Driven Development, Agile Modeling)	Assess agile process Models (Scrum, Crystal, Feature Driven Development, Agile Modeling)	Lecture	CSE3102.1	Mid Term, Quiz, Assignment & End Term
12	Software development life cycle, Assessment Model-CMMI	Details about SDLC. CMMI model	Lecture	CSE3102.1	Mid Term, Quiz, Assignment & End Term
13	Requirement Engineering, Type of requirements, Software requirements specifications	Illustrate methods of Requirement Engineering, Type of requirements, Software requirements specifications	Lecture	CSE3102.2	Mid Term, Quiz, Assignment & End Term
14	Software Engineering Practice (elicitation, analysis, documentation, validation, Management)	Explain Software Engineering Practice (elicitation, analysis, documentation, validation, Management)	Lecture	CSE3102.2	Mid Term, Quiz, Assignment & End Term
15	Size estimation- LOC Estimation, Function Count Method	Evaluate the size of project and estimation- size of project using LOC Estimation, Apply Function Count method	Lecture	CSE3102.2	Mid Term, Quiz, Assignment & End Term
16	Cost Estimation, Halstead Size Estimation	Apply Cost Estimation techniques in software projects, Evaluate software project using Halstead Size Estimation	Lecture	CSE3102.2	Mid Term, Quiz, Assignment & End Term
17	Effort Estimation-COCOMO Model	Effort Estimation- using COCOMO Model	Lecture	CSE3102.2	Mid Term, Quiz, Assignment & End Term
18	Risk Analysis and Risk Estimation	Determine Risk associated with	Lecture	CSE3102.2	Mid Term, Quiz,

		software projects			Assignment & End Term
19	Analysis Modeling: Data Modeling, Functional modeling and information flow, Data flow diagrams	Apply Analysis Modeling: Data Modeling, Functional modeling and information flow, Data flow diagrams	Lecture/Lab	CSE3102.2	Mid Term, Quiz, Assignment & End Term
20	Behavioral Modeling; The mechanics of structured analysis: Creating entity/relationship diagram	Apply Behavioral Modeling; The mechanics of structured analysis: Creating entity/relationship diagram	Lecture/Lab	CSE3102.2	Mid Term, Quiz, Assignment & End Term
21	Data flow model, control flow model, the control and process specification; The data dictionary	Design Data flow model, control flow model, the control and process specification; The data dictionary	Lecture/Lab	CSE3102.2	Mid Term, Quiz, Assignment & End Term
22	Unified modeling language – an introduction, Discussion on Class diagrams, Object diagrams, Sequence diagrams, use case diagram.	Apply Unified modeling language and develop UML Diagrams	Lecture/Lab	CSE3102.2	Mid Term, Quiz, Assignment & End Term
23	Case Study Based on Software Design	Develop Case Study Based on Software Design	Flipped Classroom	CSE3102.2	Mid Term, Quiz, Assignment & End Term
24	Design concepts and principles, the design process	Design concepts and principles and show design process	Lecture/Lab	CSE3102.2	Mid Term, Quiz, Assignment & End Term
25	Design and software quality, design principles	Relate design and software quality, design principles	Lecture/Lab	CSE3102.2	Mid Term, Quiz, Assignment & End Term
26	Design concepts: Abstraction, refinement, modularity	Recall concept of Abstraction, refinement and modularity	Lecture/Lab	CSE3102.2	Mid Term, Quiz, Assignment & End Term
27	software architecture, control hierarchy, structural partitioning, data structure, software procedure, information hiding	Develop software architecture, control hierarchy, structural partitioning, data structure, software procedure, information hiding	Lecture/Lab	CSE3102.2	Mid Term, Quiz, Assignment & End Term
28	Effective modular design: Functional independence, Cohesion, Coupling	Appraise effective modular design: Functional	Lecture/Lab	CSE3102.2	Mid Term, Quiz, Assignment

		independence, Cohesion, Coupling			& End Term
29	Software architecture: Data Design: Data modeling, data structures, databases and the data warehouse.	Develop Software architecture: Data Design: Data modeling, data structures, databases and the data warehouse.	Lecture/Lab	CSE3102.2	Mid Term, Quiz, Assignment & End Term
30	Analyzing alternative Architectural Designs, architectural complexity	Analyzing alternative Architectural Designs, architectural complexity;	Lecture/Lab	CSE3102.2	Mid Term, Quiz, Assignment & End Term
31	Mapping requirements into a software architecture, Transform flow, Transaction flow	Translate requirements into a software architecture; Recall Transform flow, Transaction flow	Lecture/Lab	CSE3102.2	Mid Term, Quiz, Assignment & End Term
32	Transform mapping: Refining the architectural design.	Apply transform mapping: Refining the architectural design.	Lecture/Lab	CSE3102.2	Mid Term, Quiz, Assignment & End Term
33	Development, Coding Standards and Conventions	Develop software systems to meet specified requirements efficiently	Lecture/Lab	CSE3102.3	Mid Term, Quiz, Assignment & End Term
34	Programing styles, code inspection, code review and walkthrough	Develop software systems to meet specified requirements efficiently	Lecture/Lab	CSE3102.3	Mid Term, Quiz, Assignment & End Term
35	Software Testing Techniques, software testing fundamentals (Software re-engineering, reverse engineering, restructuring, forward engineering).	Analyze Software Testing Techniques, software testing fundamentals	Lecture/Lab	CSE3102.4	Mid Term, Quiz, Assignment & End Term
36	Functional Testing (black Box): Boundary value analysis	Experiment with functional Testing (black Box): Boundary value analysis	Lecture/Lab	CSE3102.4	Quiz, Assignment & End Term
37	Equivalence class Testing, Decision table-based testing	Make use of Decision table-based testing	Lecture/Lab	CSE3102.4	Quiz, Assignment & End Term
38	Structural Testing (White box): Path testing	Experiment with Structural Testing (White box): Path testing	Lecture/Lab	CSE3102.4	Quiz, Assignment & End Term

39	Cyclomatic Complexity	Evaluate Cyclomatic Complexity	Lecture/Lab	CSE3102.4	Quiz, Assignment & End Term
40	Graph Matrices	Examine Graph Matrices for testing	Lecture/Lab	CSE3102.4	Quiz, Assignment & End Term
41	Data Flow Testing	Make use of Data Flow Testing	Lecture/Lab	CSE3102.4	Quiz, Assignment & End Term
42	Unit testing, Integration Testing, System Testing, Validation Testing, Testing Tools (Static Testing tools, Dynamic Testing tools, characteristics of Modern tools).	Develop test cases using different methods	Lecture/Lab	CSE3102.4	Quiz, Assignment & End Term
43	Debugging: Debugging techniques, Debugging Approaches, Debugging Tools.	Choose Debugging tools and techniques	Lecture/Lab	CSE3102.4	Quiz, Assignment & End Term
44	Software re-engineering, reverse engineering, restructuring, forward engineering.	Apply concepts of Software re-engineering, reverse engineering, restructuring, forward engineering.	Lecture/Lab	CSE3102.4	Quiz, Assignment & End Term
45	Case Study based on Software Testing	Take part in Case Study based on Software Testing	Flipped Classroom	CSE3102.4	Quiz, Assignment & End Term
46	Quality concepts, Software quality assurance, SQA activities, Software reviews	Apply Quality concepts, Software quality assurance, SQA activities, Software reviews	Lecture	CSE3102.5	Quiz, Assignment & End Term
47	Formal technical reviews: The review meeting, review reporting and record keeping, review guidelines, Formal approaches to SQA.	Recall Formal technical reviews and Formal approaches to SQA.	Lecture	CSE3102.5	Quiz, Assignment & End Term
48	Statistical software quality assurance; software reliability: Measures of reliability and availability, The ISO 9000 Quality standards: The ISO approach to quality assurance systems, The ISO 9001 standard	Explain Statistical software quality assurance and ISO 9000 Quality standards	Lecture	CSE3102.5	Quiz, Assignment & End Term
49	Characteristics of software maintenance, Software maintenance processes model	Identify characteristics of software maintenance, Software maintenance processes model	Lecture	CSE3102.5	Quiz, Assignment & End Term

50	Case Study based on Software Quality Assurance and Software Maintenance	Take part in Case Study based on Software Quality Assurance and Software Maintenance	Flipped Classroom	CSE3102.5	Quiz, Assignment & End Term
51	Introduction to DevOps	Introduce the concept and application to students	Lecture	CSE3102.5	Quiz, Assignment & End Term
52	Cloud Computing and Virtualization	Explain the concepts of Cloud Computing And Virtualization	Lecture	CSE3102.5	Quiz, Assignment & End Term
53	Migration to DevOps, DevOps Tools	Present the ideas of Migration to DevOps and DevOps Tools	Lecture	CSE3102.5	Quiz, Assignment & End Term
	Total	53 Lectures			

H. Articulation Matrix:

CO	Statement	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O 1	PS O 2	PS O 2
3102.1	Demonstrate an understanding of the fundamentals of software engineering principles, methodologies, and techniques.	1								2	1	1	3	3		
3102.2	Illustrate software requirements analysis, Software Requirements Specification (SRS) documents, software metrics, and system modeling.										1	3		3		
3102.3	Develop software systems to meet specified requirements efficiently.			3	3	3				3	1	3	3	3		
3102.4	Analyze test cases for software systems using various testing techniques.										1	1		3		
3102.5	Assess software quality, analyze evolutionary processes, and evaluate software security measures effectively.										1	2		3		

1. Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation

Faculty consultation hours: -
Depends on Faculty's free lectures hours.